Supplementary Material

Hidden Behind the Mean: Probabilistic Thermal-Sensation Diagnostics of Discomfort-Tail Risk Under Future Weather

Hongshan Guo et al.

S1 Scope

This supplementary material documents validation and reproducibility checks for the diagnostic analyses in the main manuscript. It does not report closed-loop control, energy savings, demand-response performance, or equipment-facing outcomes.

S2 Run Inventory and Trace Integrity

The final diagnostic run contains 144 annual DOE Medium Office cases: six cities, two emissions scenarios, four time slices, and three annual weather-file severities. Each annual trace contains 35,040 15-minute rows and 16,704 occupied rows. The zone-raw trace schema contains 45 raw environmental fields: air temperature, mean radiant temperature, and relative humidity for each of 15 conditioned zones.

The final zone-raw diagnostic run produced 144 per-case trace files, 144 EnergyPlus error files, and 144 EnergyPlus end files. All annual cases completed without fatal EnergyPlus errors. Sixteen cases reported warmup load-convergence Severe messages; these cases completed successfully and are retained with this quality-assurance note.

S3 Weather Panel and Case Provenance

The weather panel is used as a deterministic stress-test panel, not as a probabilistic climate ensemble. The 144 cases are the Cartesian product of six cities (Ahmedabad, Beijing, Guangzhou, Houston, Kolkata, and Phoenix), two emissions scenarios (SSP2-4.5 and SSP5-8.5), four time slices (baseline 2020s, near 2030s, mid 2050s, and late 2080s), and three selected annual severities (typical, hot, and heatwave-extreme). The three annual severities are selection labels within the local weather-file generation workflow; they should not be read as equal-probability climate outcomes.

The public reproducibility repository archives the run manifest, weather-file provenance table, case identifiers, trace-schema checks, and checksum records needed to audit the mapping between manuscript cases and generated annual weather files. Large weather files and per-timestep traces may be represented by checksums rather than redistributed directly where file size or third-party restrictions apply.

S4 Metabolic Profiles

The metabolic-spread diagnostic reuses the same simulated environmental traces and re-evaluates the TSV probability model under six metabolic-load profiles: 90.0, 93.2, 100.0, 108.0, 110.0, and 120.0 W/person. Under the convention used in the manuscript, 120 W/person corresponds to 1.2 met, so the six profiles correspond to 0.900, 0.932, 1.000, 1.080, 1.100, and 1.200 met. These profiles are sensitivity cases for the reported-TSV model. EnergyPlus was not rerun with altered internal heat gains, and the profiles are not demographic strata assigned to simulated occupants.

S5 Held-Out TSV Prediction Validation

The primary TSV predictor and the no-PMV robustness predictor were evaluated on the same held-out test partition of 22,223 ASHRAE Global Thermal Comfort Database II records. The split was stratified over the seven rounded TSV classes. The rounded-PMV baseline rounds PMV to the nearest TSV class and clips the value to $[-3, +3]$; it is a deterministic scalar-label baseline, not a calibrated probability model.

Table S1: Held-out TSV validation for the primary predictor, no-PMV robustness predictor, and rounded-PMV class baseline.

Predictor	Exact (%)	Within ±1 (%)	MAE	Tail recall (%)	Tail F1 (%)	Log loss	Tail MAE
Primary TSV model	47.3	84.1	0.73	23.4	34.0	1.454	0.277
No-PMV TSV model	46.9	83.8	0.73	23.1	33.5	1.476	0.278
Rounded PMV class	33.2	76.5	0.97	22.8	26.6	–	0.258

Table S2: Observed class support and class-specific recall on the held-out test partition. Values are percentages except support counts.

Predictor	TSV -3	TSV -2	TSV -1	TSV 0	TSV +1	TSV +2	TSV +3
Support (n)	443	1238	3755	9834	4064	1982	907
Primary TSV model	25.5	6.9	12.4	88.0	16.2	15.7	23.9
No-PMV TSV model	23.7	6.5	13.0	87.4	15.4	15.8	23.2
Rounded PMV class	13.8	10.4	27.6	47.8	26.3	14.5	10.5

S6 Calibrated Tail-Probability Comparison

As a scalar-reference check, PMV was also given a calibrated probability baseline. This baseline used only |PMV| as the input and the binary tail label $\mathbf{1}\{\text{TSV} \leq -2 \text{ or TSV} \geq +2\}$ as the target. An isotonic regression was fitted on the non-test partition and evaluated on the same held-out test records used in Tables S1 and S2. This is not an ordinal TSV model; it only tests how much severe-tail probability can be recovered from PMV alone after calibration. Table S3 compares that scalar baseline with the primary and no-PMV TSV probability models using the same tail-probability metrics.

Table S3: Held-out tail-probability comparison for the TSV probability models and the PMV-only calibrated scalar baseline. F1 is computed by screening each predicted tail probability at $p_{\text{tail}} \geq 0.20$.

Predictor	n	Mean predicted	Observed	Brier	ECE	MCE	F1 (%)	AUROC	Avg. precision
Primary TSV model	22223	0.207	0.206	0.137	0.011	0.047	47.1	0.747	0.487
No-PMV TSV model	22223	0.207	0.206	0.138	0.006	0.012	46.4	0.747	0.483
PMV-only calibrated tail baseline	22223	0.206	0.206	0.161	0.003	0.017	32.6	0.571	0.260

The PMV-only baseline matched the overall tail prevalence after calibration, with mean predicted $p_{\text{tail}} = 0.206$ and observed tail frequency 0.206. Its calibration error was low, but its discrimination was limited: AUROC was 0.571 and average precision was 0.260, compared with AUROC 0.747 and average precision 0.483–0.487 for the TSV probability models. This supports the limited interpretation in the main text: PMV contains useful scalar information, but PMV alone is not the same diagnostic object as a calibrated TSV probability distribution.

S7 Contributor-Grouped Robustness Check

The record-level split above tests held-out records under the same overall ASHRAE-derived data distribution. As a stricter robustness check, the ordinal predictors were also re-estimated under five contributor-grouped train/calibration/test splits. Contributor groups were kept disjoint across training, calibration, and test partitions. Each split used 43 contributor groups for training, 10 for calibration, and 10 for testing; the mean test size was 26,484 records.

This grouped check is not used as the main model-selection criterion because contributor groups differ substantially in geography, building type, measurement protocol, and class balance. It is included to expose domain-transfer sensitivity. As expected, performance was lower than under the record-level split, especially for tail-class discrimination, but the comparison between the primary and no-PMV feature sets remained qualitatively similar.

Table S4: Contributor-grouped validation over five disjoint-group splits. Values are mean (standard deviation) across splits.

Feature set	Exact (%)	Within ±1 (%)	Class MAE	Tail F1 (%)	Log loss	Tail ECE
Primary	42.3 (3.0)	80.1 (2.2)	0.825 (0.047)	6.1 (5.1)	1.745 (0.222)	0.060 (0.026)
No-PMV	42.0 (3.2)	79.4 (1.7)	0.838 (0.042)	4.8 (3.3)	1.804 (0.275)	0.057 (0.026)

S8 Discomfort-Tail Probability Calibration

The central probability variable in the manuscript is

$$p_{\text{tail}} = \Pr(\text{TSV} \leq -2) + \Pr(\text{TSV} \geq +2). \tag{1}$$

To check whether this value behaves as a calibrated probability on held-out data, predicted p_{tail} values were grouped into fixed probability bins. Within each bin, the mean predicted p_{tail} was compared with the observed frequency of tail-class records. Tail expected calibration error (ECE) is the bin-count-weighted mean absolute difference between predicted and observed frequency. Tail maximum calibration error (MCE) is the largest bin-level absolute difference.

Table S5: Held-out calibration summary for predicted discomfort-tail probability.

Predictor	n	Mean predicted	Observed	Tail Brier	Tail ECE	Tail MCE
Primary TSV model	22223	0.207	0.206	0.137	0.011	0.047
No-PMV TSV model	22223	0.207	0.206	0.138	0.006	0.012

Table S6: Primary TSV model: fixed-bin held-out calibration for p_{tail}.

Bin	n	Mean predicted	Observed	Abs. error
0.00–0.05	1024	0.036	0.027	0.009
0.05–0.10	4021	0.077	0.072	0.005
0.10–0.15	4652	0.124	0.115	0.009
0.15–0.20	4251	0.175	0.163	0.012
0.20–0.30	4277	0.244	0.253	0.009
0.30–0.40	2005	0.350	0.359	0.009
0.40–0.60	919	0.469	0.516	0.047
0.60–1.00	1074	0.713	0.701	0.012

The held-out calibration check supports using p_{tail} as a diagnostic probability rather than only as an uncalibrated score. The primary model's mean predicted p_{tail} was 0.207, close to the observed held-out tail frequency of 0.206. Fixed-bin ECE was 0.011 for the primary predictor and 0.006 for the

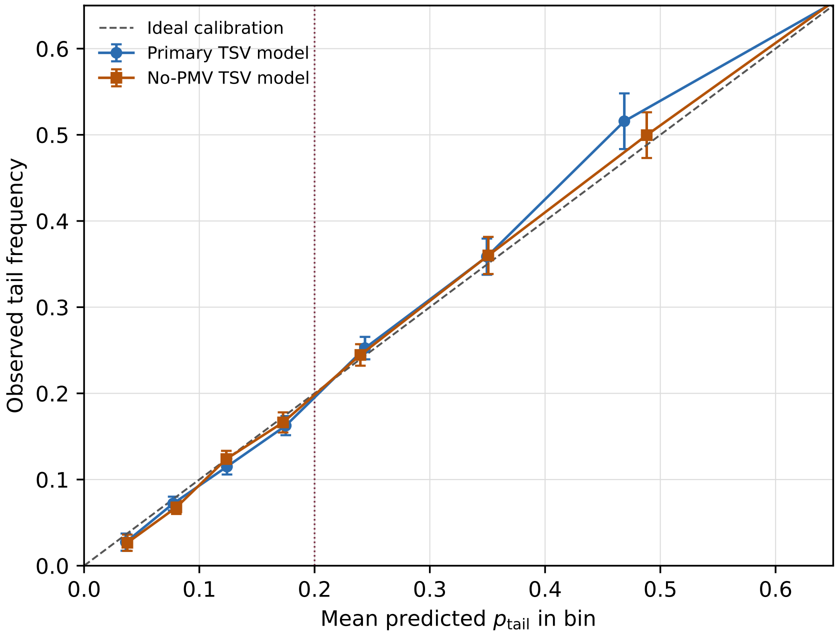

Figure S1: Held-out reliability check for predicted discomfort-tail probability. Points show fixed bins of predicted p_{tail}; vertical bars show approximate 95% binomial intervals for the observed tail frequency in each bin. The dashed line marks ideal calibration, and the dotted vertical line marks the manuscript's reporting screen $p_{\text{tail}} = 0.20$.

no-PMV predictor. These values do not remove the sparse-tail limitation noted in the manuscript, but they show that the aggregate tail probability is well aligned with observed tail frequency on the held-out record-level split.

S9 Generated Diagnostic Outputs

The reproducibility workspace contains the generated outputs used by the manuscript and this supplement:

- held-out TSV validation summaries and class-recall tables;
- contributor-grouped validation summaries for primary and no-PMV feature sets;
- p_{tail} calibration summaries, fixed-bin reliability tables, and reliability plots;
- scalar PMV and expected-TSV comparison diagnostics;
- threshold-sensitivity summaries for p_{tail} screens from 0.05 to 0.40;
- zone-aggregation summaries, zone-size mosaic inputs, and floor-position diagnostics;
- metabolic-spread summaries and zone-resolved metabolic-profile diagnostics;
- EnergyPlus trace-schema and run-quality audits.

Large per-timestep traces and third-party ASHRAE Global Thermal Comfort Database II records are not redistributed directly. The submission repository should provide scripts, processed summaries, model artifacts, weather-file provenance, and checksums for auditing the reported tables and figures where redistribution is permitted.